

1 MES-Cockpit - Production Monitoring

1.1 General

The production monitoring provides the user with an system-independent status overview for the following objects:

- Machines/workplaces
- Running operations
- Registered orders
- Downtime hit list
- Produced quantities
- KPI diagram



Please note: Data of the online overview are updated in a cyclic interval every three minutes by the connected HYDRA systems:

1.2 Selection criteria

The following selection criteria are provided in selection lists:

- Workplaces/ machines
- Machine groups
- Cost centers
- Article/item (restricts the displayed operations)



Please note: There are no selection criteria to narrow down time periods as the displayed data refers to the current shift of the objects.

1.3 Objects of online overview

1.3.1 Machines/workplaces

The overview of machines/workplaces provides current information about workplaces (in table form) for which the registered user is authorized by the responsibility area and that match the entered selection criteria.

This information is shown for workplaces:

- Machine number
- Machine name
- Cost center
- Machine group
- Site
- Current status (status text, status color and number)
- Status since
- Duration so far
- Target cycle of machine
- Actual cycle of machine
- Produced yield and scrap quantity of the machine of the current shift
- Actual cycles of machine
- Actual cycles (yield) of machine

1.3.2 Running operations

The overview of running operations shows the currently registered operations in table form. Filters are set as follows:

- Only those operations are shown that are logged on to a workplace displayed in the overview of workplaces. Therefore, if the displayed data is filtered by workplaces, the list of operations will also be affected.

The following information is shown for operations:

- MES order number
- Article number
- Article description
- Order number
- Site
- Date/time of logon
- OP description
- Target quantity
- Yield quantity
- Scrap
- Unit (P)
- Workplace
- First logon
- Last logoff
- Last interruption

- Planned start
- Planned end
- Earliest start
- Latest end
- Scheduled start
- Scheduled end
- Total setup time
- Processing time



Please note: If an OP is logged on to two machines at the same time, it will only be shown once in the list. But if filtering is based on machines, it will be shown for both machines.

1.3.3 Registered orders

The overview of registered orders represents in table form information on order headers for which at least one operation is currently logged on. Filters are set as follows:

- Therefore, if the displayed data is filtered by operations, the list of orders will also be affected.

The following information is shown for orders:

- Order number
- Finished article
- Article description
- Sales order
- Site
- Current status (status text and colors)
- Status since
- Basic start date
- Basic end date
- Scheduled start
- Scheduled end Target quantity (B)
- Yield (B)
- Scrap (B)
- Planned lead time
- Target setup time
- Target processing time
- Target labor utilization

- Target execution time
- Retention period of order
- Lead time
- Setup time
- Processing time
- Downtime
- Assignment time
- Labor utilization

1.3.4 Downtime hit list

A bar chart shows the malfunctions that have occurred in the current shift providing information about the duration and number of workplaces displayed and/or selected in the overview.

1.3.5 Produced quantities

A bar chart shows the produced yield and scrap quantities of the current shift for each cost center.

1.3.6 KPI diagram

The KPI diagram shows the following key performance indicators for the current shift and all selected and/or displayed workplaces:

- Rate of capacity utilization
- Setup rate
- Scrap rate

The specified formula of formula management is used as basis for calculation.